Lianzijun (Lian) Wang

Department of Biology, Duke University

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EDUCATION

Duke University Durham, NC

Ph.D. in Biology, Department of Biology

In progress

Baylor University Waco, TX

Bachelor of Science, Biochemistry and Biology, Cell & Molecular Biology concentration

GPA: 4.0; Summa Cum Laude

RESEARCH EXPERIENCE

Department of Biology, Duke University 2023 - Present

Durham, NC

Ph.D. Researcher; Mentor: Dave Sherwood, PhD

- Investigate how invasive cells breach basement membrane using the C. elegans anchor cell invasion model, with emphasis on protease-independent invasion and physical basement membrane displacement.
- Study how UNC-52/perlecan regulates basement membrane architecture, mechanics, and plasma membrane protection during anchor cell invasion.
- Examine how MMP-deficient animals adapt mitochondrial network morphology and metabolism to support invasion through a mechanically challenging basement membrane environment.
- Analyze how the spectrin cytoskeleton supports invasive protrusion formation, plasma membrane stability, and mechanical resilience during basement membrane breaching.

Department of Biology, Baylor University March 2021 - May 2023

Waco, TX

Undergraduate Researcher; Mentor: Myeongwoo Lee, PhD

- Used CRISPR-Cas9 genome editing and C. elegans microinjection to generate and characterize alleles affecting extracellular matrix and cytoskeletal genes.
- Generated and analyzed mutant alleles in pat-3/beta integrin, unc-52/perlecan, cle-1/collagen XVIII, sned-1, dxbp-1, deb-1/vinculin, lon-1, lam-3/laminin alpha, and mec-9.
- Investigated extracellular matrix-cell adhesion mechanisms regulating C. elegans development, mechanosensation, egg-laying, and muscle function.

B-TRUE Summer Fellow Program, Baylor University Summer 2020

Baylor Transdisciplinary Research Undergraduate Experience Program, Waco, TX

- Analyzed and presented a natural product synthesis study using pharmacophore-directed retrosynthesis to generate rameswaralide-inspired Sinularia tricyclic cores and evaluate their bioactivity.

PUBLICATIONS

Peer-reviewed publications

Kenny-Ganzert, I. W., Basta, L. P., **Wang, L.**, Chi, Q., Kelley, L. C., Su, C. Y., Leyhr, J., Morton, K. S., Meyer, J. N., & Sherwood, D. R. **(2026)**. Specialized high-capacity mitochondria fuel cell invasion. Current Biology, 36(8), 1959-1976.e10. doi:10.1016/j.cub.2026.03.023. PMID: 40654879.

Bumm, J. H., Lee, D. D., Qiu, Z., Choi, A. H., Kolluri, M., Olsen, M. J., **Wang, L.**, & Lee, M. **(2025)**. The beta integrin modulates serotonin sensitivity via NPxY motifs to regulate egg laying and mechanosensation behaviors in Caenorhabditis elegans. Developmental Dynamics. Advance online publication. doi:10.1002/dvdy.70090. PMID: 41117756.

Serrano, M. V., Cottier, S., **Wang, L.**, Moreira-Antepara, S., Nzessi, A., Liu, Z., Williams, B., Lee, M., Schneiter, R., & Liu, J. **(2025)**. The C. elegans LON-1 protein requires its CAP domain for function in regulating body size and BMP signaling. Genetics, 229(2), iyae202. doi:10.1093/genetics/iyae202. PMID: 39657262.

Sadeghian, F., Ibrahim, I., Ravichandran, L., Henderson, G., Acharya, A., **Wang, L.**, Lee, M., & Cram, E. J. **(2023)**. An integrin binding motif in TLN-1/talin plays a minor role in motility and ovulation. microPublication Biology, 2023. doi:10.17912/micropub.biology.000726. PMID: 36685729.

Qiu, Z., Park, A., **Wang, L.**, Wilsey, R., & Lee, M. **(2022)**. The RGD (Arg-Gly-Asp) is a potential cell-binding motif of UNC-52/PERLECAN. Biochemical and Biophysical Research Communications, 586, 143-149. doi:10.1016/j.bbrc.2021.11.083. PMID: 34844120.

Wang, L., Qiu, Z., & Lee, M. **(2021)**. Mutations in the cell-binding motif of lam-3/laminin alpha reveal hypercontraction behavior and defective sensitivity to levamisole in Caenorhabditis elegans. microPublication Biology, 2021. doi:10.17912/micropub.biology.000485. PMID: 34723150.

Preprints

d'Humieres, J., **Wang, L.**, Sherwood, D. R., & Plastino, J. **(2025)**. The actin protrusion deforms the nucleus during invasion through basement membrane. bioRxiv. doi:10.1101/2025.03.13.643012. PMID: 40161654.

PRESENTATIONS

Wang, L. (2026). Triangle Cytoskeleton Meeting (TCM). Selected for oral presentation.

Wang, L. (2024). Plasma membrane damage and repair during anchor cell invasion. Triangle Cytoskeleton Meeting. Poster presentation. Authors: Lianzijun (Lian) Wang, Kieop Park, Isabella Antonio, Dave Sherwood.

Wang, L. (2023). DEB-1/Vinculin proline-rich-linker (PRL) plays important roles in touch sensitivity and egg-laying of Caenorhabditis elegans. Texas Genetics Society, 50th Annual Meeting, Austin, TX. Poster presentation.

Wang, L. (2022). Mutations in the cell-binding motif of lam-3/laminin alpha. Capital of Texas Undergraduate Research Conference, University of Texas at Austin. Oral presentation.

Wang, L. (2022). Mutations in the cell-binding motif of lam-3/laminin alpha reveal hypercontraction behavior and defective sensitivity to levamisole in Caenorhabditis elegans. CURES Biology Symposium, Baylor University. Poster presentation.

HONORS AND AWARDS

URSA Scholars Week 2023 Outstanding Platform Presentation April 2023
Awarded for the presentation "DEB-1/Vinculin proline-rich-linker (PRL) plays important roles in touch sensitivity and egg-laying of Caenorhabditis elegans."

Folmar Award June 2022 - May 2023
Research grant from Baylor University's Jack G. and Noma Jean Folmar Research Fund supporting CRISPR-Cas9 characterization of LON-1/CRISP functions.

URSA Mini-Research Grant March 2022 - May 2022
Undergraduate Research and Scholarly Achievement program grant supporting student wages.

URSA Scholars Week 2022 Outstanding Platform Presentation April 2022
Awarded for the presentation on lam-3/laminin alpha cell-binding motif mutants.

Chemistry Honor Society 2021 - 2023
Member, Baylor University Chemistry Honor Society.

Provost's Gold Scholarship 2019 - 2023
Scholarship awarded by Baylor University on the basis of academic success.

Dean's Academic Honor List 2019 - 2023

TEACHING AND LEADERSHIP EXPERIENCE

Department of Biology, Duke University Fall 2024 - Spring 2025
Teaching Assistant, BIO 220 Cell Biology and BIO 202 Genetics and Evolution
- Led TA review sessions and office hours to support student learning; graded assignments and exams.

Learning Assistant, Senior Biology Laboratory January 2022 - March 2023
BIO 4108 Gene and Development, Baylor University
- Assisted with scheduled class administration, laboratory material preparation, experimental protocol writing, and coordination of laboratory activities.

Supplemental Instruction Leader August 2020 - December 2021
BIO 1305 and BIO 1306, Baylor University
- Hosted and led 90-minute-long review sessions twice a week for introductory biology courses with a class size of 80 students.
- Designed and distributed questions and study guides for the courses.

SERVICE AND COMMUNITY ENGAGEMENT

Duke University Hospital, Music Program 2026–Present
Volunteer Musician
- Perform live music through the Duke Hospital music volunteer program for patients, families, and hospital staff.